

Syllabus

Welcome to *Data+AI*, Spring '26

This site available at: go.uic.edu/cs218

CS 294, CRN 48392, hosted by Chicago Tech Circle





The temporary Spring course registration name is CS 294 (until CS 210 gets approved), with CRN 40392. For College of Business students only, the course is cross-listed as FIN 294.

The course goals are for you to:

1. Develop data and AI skills applicable to your own areas of study
2. Become more marketable

What AI Says About the Course:

- Watch a [2-minute video summary](#) of what students learn, or hear a student-focused [5-minute podcast summary](#) of the course. Both of these were generated by AI (NotebookLM), from the course materials.
- See [cartoons](#) student created in the course, with the help of ChatGPT.

See the [course schedule](#) for a more detailed breakdown of topics covered.

1. Leveraging AI Effectively and Ethically:

Designing, refining, and critically evaluating AI prompts to generate useful outputs (e.g., tutorials, code, descriptive text) while recognizing and mitigating limitations such as algorithmic bias, attribution issues, and hallucinations.

2. Predictive Modeling and Machine Learning:

AI-assisted use of library-based machine learning models, including proper data partitioning and the application of Machine Learning (ML) algorithms (e.g. regression and decision trees) to solve real-world problems.

3. Quantitative and Statistical Reasoning:

AI-assisted application of basic statistical methods (e.g., calculating descriptive metrics, understanding variance, range, and quartiles) to interpret data, differentiate between correlation and causation, and support sound data-driven conclusions.

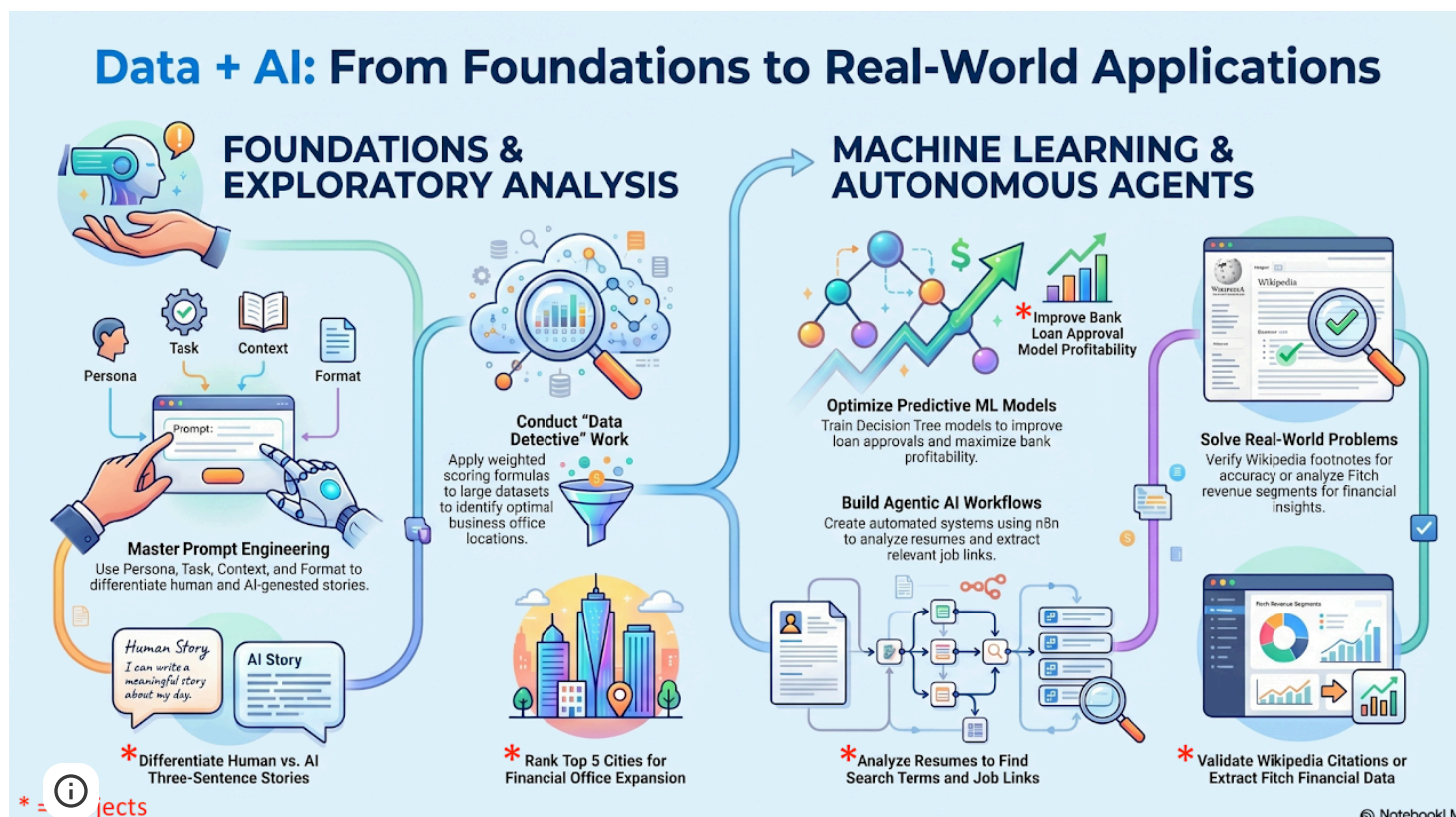
4. Foundational Data Handling and Analysis:

Using basic tools—from spreadsheets to programming—to combine, clean, organize, transform, and explore data. This includes practical skills in managing everyday datasets and recognizing tool limitations.

5. Data Communication and Value Creation:

Translating analytical insights into clear, compelling visual stories, integrating these insights into business contexts to create value and drive data-informed decision-making.

The above content can be summarized by this NotebookLM-generated infographic:



Instructors:

1. Dale Reed

email (remove spaces): reed @ uic.edu

Web: bit.ly/dalereed is a shortcut to sites.google.com/site/dalereed

Office: CDRLC 4431 (the new CS building). See [office drop-in times](#).

2. Baker Franke

email (remove spaces): bfranke @ uic.edu

[UIC News article](#)

3. Tomas Malik (Instructor and grad TA) email (remove spaces): tmalik6 @ uic.edu

TAs are listed on the [CS 218 TA & Drop-in Times page](#). Grad and undergrad TA hours will be held in CDRLC 2441. We're also available immediately after class, or by appointment.

Prerequisites: 1. Basic math, 2. At least 30 credit hours into your major, 3. A GPA of at least 2.75

This course is intentionally designed for *all* UIC students. You don't need to have a technical background to take this course, though we assume you know the basics of how to use a Mac or Windows computer.

For Finance students, you must also have a major, minor, or concentration in one of the College of Business areas of study.

There is no textbook for this course, though there will be regular assigned readings and activities associated with each class.

You will need:

1. A **laptop computer**, which you should bring to every class. We will use these the first class!
2. Your **[UIC Google Account Set up](#)**. We serve almost all documents for the course through Google Suite products (slides, docs, sheets). Much of your work will be submitted via Google forms. You may already have an existing google account (that you setup without realizing it). Please make sure you have at least a few Gigabytes of storage space free in your Google drive.
3. Take **[this pre-class survey](#)**. It consists of 27 multiple choice questions and will take ~7 minutes. While everyone has to do it, you can choose whether or not you want your responses to be part of the study. Many of the questions ask how familiar you are with some of the things we'll be working on this semester, though we don't assume that you know any of it! We're doing this survey to see how well this course goes and to report out to Google, who helped fund the development of this course.
4. You should have gotten an email invitation to join **Piazza**, the online discussion board we will be using for this class. Rather than emailing questions to course instructors, post your questions on Piazza, which is specifically designed to get you help quickly and efficiently from classmates, the TA, and instructors. We really do want to talk to you and want you to engage in conversation with each other. We believe each of us has perspectives and insights to share that can help others in the class, regardless of starting point or level of expertise. Seeking help is normal and expected. We encourage you to ask any question, but recognize that sometimes this is embarrassing, which is why your posts can be anonymous to other students.
To access Piazza (or sign up if you didn't get the invite), follow the link in the navigation bar at **<https://piazza.com/uic/spring2026/cs294akacs218>**.
5. A **subscription to ChatGPT Plus** at **chatgpt.com/plans/plus**, which is \$20/month (price as of Jan '26).
6. We will be using an online environment called **[Google Colab](#)**. Try the free version first. Depending on the performance you get from the free version, you *may* later need to purchase Pay-as-you-Go **[Compute Units](#)**, which is \$10 / 100 compute units (price as of Aug '25).
7. In the last 6-8 weeks of the semester, we will be using the "Starter" version of an industry agentic AI tool called n8n (pronounced "en-eight-en"). You get two weeks for a free trial, and then it is **[\\$24 a month](#)**. We will *try* to use only one month's worth, but it is possible that we will need two months' worth, as it will be used for the final project. So, don't sign up for this now and wait until we cover that part of the course. However, **do** plan on needing to spend \$24-\$48 for it. We are also exploring the option of performing our own local installation, although that is not certain.
8. **Microsoft Excel spreadsheet** program. This is available for free to UIC students from the **[University of Illinois Webstore](#)** as part of Microsoft 365.

There is regular homework, regular quizzes, and significant group projects, but there are no exams.

Semester final grades will be credit / no credit for the course. To receive credit for the class you will need to earn 80% or higher. The course is designed so that this is readily doable if you always show up to class and do the regularly assigned work. Grades will be updated regularly and displayed on the course website so you know where you stand in the course.

10% Surveys. We will have a brief survey each week. We will also have a survey before the first week, and at the end of the course.

20% Quizzes. We will usually have a brief quiz every day based on the homework or other assigned activity. The lowest quiz grade will be dropped.

10% In-class work. We will often collaboratively edit google slides together in class, and sometimes give presentations

20% Homework. You will submit homework using a google form. Homework will typically be graded on a 0..3 scale as follows: 0 - Did not do, 1- Significant components missing, 2 - Complete, 3 - Far exceeds expectations (extra credit). The lowest homework grade will be dropped.

40% Projects. Each project will have an individual component (50%) followed by a combined group component (50%)

8% Project 1

8% Project 2

8% Project 3

16% Final Project (#4). Instructors will determine near the end of the semester whether this will be group or individual work.

Using Generative AI (e.g. ChatGPT)

We will be using Generative AI (e.g. ChatGPT) extensively for just about everything in this course. (Usually we will abbreviate Generative AI to just be GenAI.) To receive credit you will be often be asked to *reflect upon* and capture your *process* of using GenAI. **Not using Generative AI will likely result in you do poorly in this class!** You will also be asked to post insights and advice to your classmates on how to get the most out of Generative AI.

Late Penalties

No late work is accepted because this class is oriented around in-person discussion, group work, and

the instructor about this within one week of the time the grades for the work in question are first posted for the class.

Class Norms



Here are the class norms, suggested by the class, that we are adopting for this semester. [This google doc](#) has all the original suggestions, with our comments on them.

1. Make space / take space, with everyone participating.
2. Be respectful and patient in speaking and listening.
3. Keep an open mind to new people and ideas
4. Don't be afraid to ask questions.
5. Make course materials as easy to use as possible

Note that the CS Department also has a [Code of Conduct](#) of reasonable behavior we will all expect of each other.

Logistics (Missing class, LOA, Incompletes, Regrades)



The vast majority of students do their own work, as we expect you to do. We encourage conversation and collaboration, as detailed below indicating what you *may* and *may not* do in collaboration with others.

Academic dishonesty is anything that gives you an unfair academic advantage.

We believe you are capable of learning the concepts in the course and doing the work, and are committed to helping you get there. If you are struggling in the course please ask us for help.

As already mentioned, for this class it is assumed you will use Generative AI (e.g. ChatGPT) extensively for just about everything. Give attribution when using someone else's work and do not represent it as your own. For this course likely most of your work should include a statement such as "Generative AI was used to help in the creation of this work."

Class

Do

1. Talk to people around you during pair-share discussion
2. Use Generative AI to get your questions answered and to generate content.

Don't

1. Enter content for someone else to give them quiz points or collaborative google slide entries
2. Communicate with someone else in class during quizzes

Projects

Do

1. Discuss conceptual solutions to projects, both in-person and online
2. Show people your code in-person, and describe it (but no electronic copies!)
3. Suggest prompts to help others generate the content they need
5. Post and discuss your solutions
6. Use Generative AI to generate code, suggest ideas, and create content.

Don't

1. Send or post an electronic copy of a draft or finished version of your project or assignment to students outside your group before the deadline
2. Represent someone else's work as your own.

4. Turn in the GenAI product as your own without giving attribution

Any student who fails to follow the academic honesty policy on any course work will get a grade of 0 on that work and will be dropped 10% from their final course average, for each offense. This penalty applies to all prohibited behavior listed above in the *Don't* categories, unless specifically allowed by the instructor.

Academic dishonesty will result in an Academic Integrity Incident Report go.uic.edu/academicIR going to the UIC Office of the Dean of Students. Details are given on the [Academic Integrity page](#), which includes a link to the [Student Disciplinary Policy](#).

Acknowledgments



Land Acknowledgment

At UIC we build on the work of others. We want to acknowledge that the land upon which we live and work was forcibly taken from Native American people. As described [here](#), "The Chicago area is located on ancestral lands of indigenous tribes, such as the Council of the Three Fires--comprised of the Ojibwe, Odawa, and Potawatomi Nations--as well as the Miami, Ho-Chunk, Menominee, Sac, Fox, Kickapoo, and Illinois Nations." See the [Invasion of America](#) video (90 sec) showing how Native American peoples lost their land between 1776 and 1887.



CHICAGO **techcircle**

A large red circle containing the white text "uic" in a bold, sans-serif font.

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